Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

# ENVIRONMENTAL SCIENCE

## Paper 2 (Generated practice paper - Set 24)

Assessed: The living environment, Biological resources, Sustainability, Research methods

**Time allowed: 3 hours**

### Materials

For this paper you may use:
- a calculator.

### Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

### Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

**For Examiner's Use**

| Question | Mark |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |
| 6 | |
| 7 | |
| 8 | |
| 9 | |
| 10 | |
| 11 | |
| TOTAL | |

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**Answer all questions in the spaces provided.**

**0 1** Table 1 shows climate data for two sites where a farmer is considering growing maize, a frost-sensitive crop that needs a thermal growing season of at least 120 days.

**Table 1**

| Feature | Site A: valley floor | Site B: south-facing slope |
|---|---|---|
| Frost-free period / days | 110 | 150 |
| Days with mean temperature above 5 °C | 160 | 175 |
| Mean summer rainfall / mm | 180 | 150 |
| Gradient / % | 1 | 8 |

**0 1 . 1** Use Table 1 to explain which site is more suitable for maize.

**[3 marks]**

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**0 1 . 2** Describe **three** ways in which temperature can be controlled to increase crop productivity.

**[3 marks]**

**1** .....

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**2** .....

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**3** .....

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**0 1 . 3** Explain why keeping livestock warm increases the amount of food converted to growth.

**[2 marks]**

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**0 1 . 4** Explain how day length affects livestock production and how it can be manipulated.

**[2 marks]**

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**10**

**0 2 . 1** Which feature of the Earth prevents most atmospheric gases from escaping into space? Tick **one** box.

**[1 mark]**

A Its mass ☐

B Its magnetic field ☐

C Its distance from the Sun ☐

D Its speed of rotation ☐

**0 2 . 2** Explain how the speed of rotation of the Earth and the tilt of its axis affect the conditions for life.

**[2 marks]**

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**0 2 . 3** Explain why the presence of liquid water on Earth depends on **both** the distance from the Sun **and** atmospheric pressure.

**[2 marks]**

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**0 3 . 1** Explain why an electronic meter used to measure dissolved oxygen must be calibrated, and what should be done if the reading cannot be adjusted.

**[2 marks]**

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**0 3 . 2** Give **three** types of indirect evidence of the presence of an animal and the information each can provide.

**[3 marks]**

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**2** .....

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**3** .....

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**0 3 . 3** Explain how auditory monitoring can be used to study bats.

**[2 marks]**

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**0 3 . 4** Explain **three** assumptions of the mark-release-recapture method that may not be met.

**[3 marks]**

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**0 4 . 1** Compare selective breeding, asexual reproduction and cross-breeding as methods of improving livestock or crops.

**[6 marks]**

**0 4 . 2** Explain how artificial insemination and embryo transfer increase the rate at which desirable characteristics spread through a livestock population.

**[4 marks]**

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**0 5 . 1** Describe how each of the following abiotic factors can be managed to increase biodiversity in a nature reserve.

**[6 marks]**

**Light**

**Dissolved oxygen**

**Nutrients**

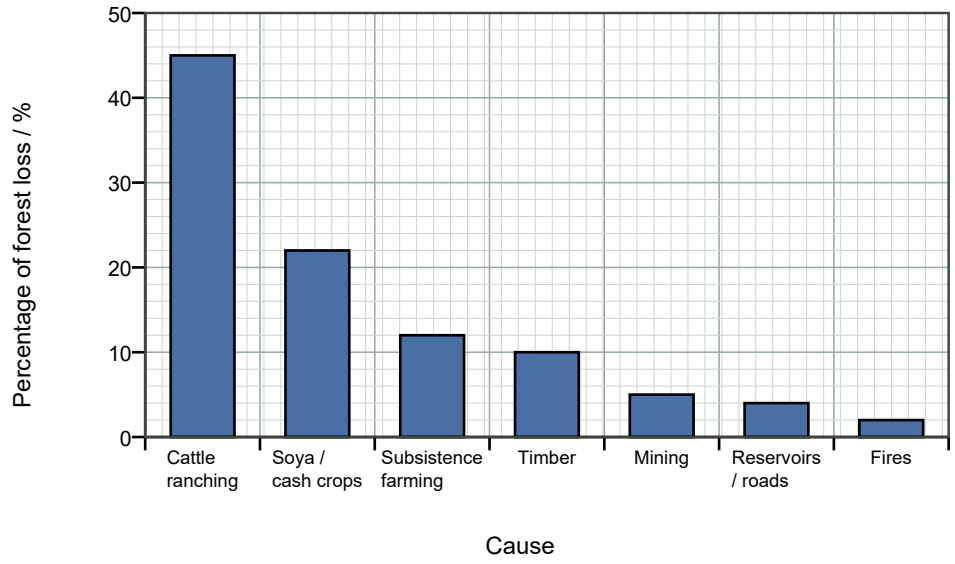
**0 5 . 2** Explain how biotic factors can be managed to increase the survival of a species. Give **four** examples.

**[4 marks]**

10

**0 6** Figure 1 shows the main causes of tropical rainforest loss in a South American country between 2000 and 2020.

**Figure 1**




**0 6 . 1** Explain why traditional 'slash and burn' subsistence agriculture was sustainable but has become a threat.

**[3 marks]**

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**0 6 . 2** Explain why clearance for cattle ranching or cash crops is usually unsustainable.

**[2 marks]**

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**0 6 . 3** Describe how each of the following threatens rainforest.

**[3 marks]**

**Mineral extraction** .....

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**HEP reservoirs** .....

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**Climate change** .....

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**0 6 . 4** Describe how a debt-for-nature swap works.

**[2 marks]**

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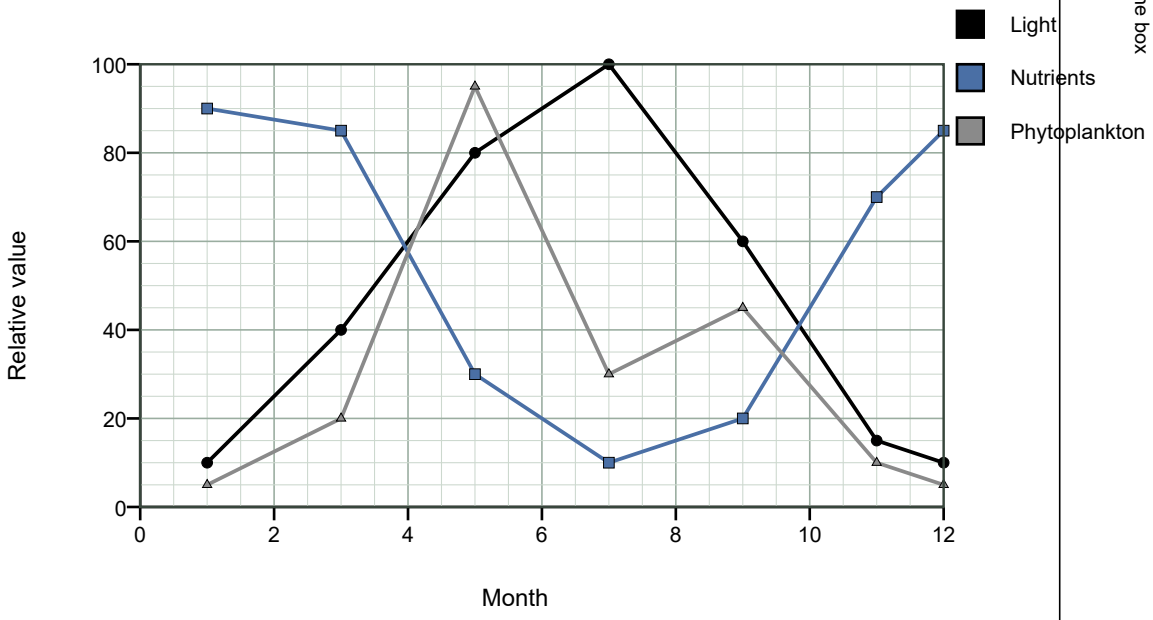
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**10**

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**0 7** Figure 1 shows the seasonal changes in light intensity, nutrient concentration and phytoplankton biomass in the surface water of a temperate ocean in the northern hemisphere.

**Figure 1**




**0 7 . 1** Use Figure 1 to explain the spring peak and summer decline in phytoplankton biomass.

**[4 marks]**

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**0 7 . 2** Explain why tropical open oceans have low productivity all year despite high light and temperature.

**[2 marks]**

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**0 7 . 3** Explain why areas of upwelling and coastal waters are the most productive parts of the sea.

**[3 marks]**

**0 7 . 4** Explain why phosphate is often the limiting nutrient in the sea.

**[1 mark]**

**10**

**0 8 . 1** Describe how agriculture alters the hydrological cycle.

**[3 marks]**

**0 8 . 2** Explain how each of the following consumer choices can reduce the environmental impact of food production.

**[4 marks]**

**Seasonal / local food** .....

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**Organic food** .....

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**0 8 . 3** Explain why post-war European governments introduced grants and guaranteed prices, and describe **one** environmental consequence.

**[3 marks]**

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**0 9** Table 1 lists some techniques used in ecological monitoring.

**0 9 . 1** Complete Table 1 by giving one application and one limitation of each technique.

**[6 marks]**

**Table 1**

| Technique | Application | Limitation | |
|---|---|---|---|
| Image databases of unique markings | | | |
| eDNA sampling | | | |
| Geolocator tags | | | |

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**0 9 . 2** Explain how satellite radar (microwave) sensors are useful for monitoring oil pollution and sea conditions.

**[2 marks]**

**0 9 . 3** Give **two** types of indirect evidence of an animal's presence and the information each can give.

**[2 marks]**

**1**

**2**

**10**

**1 0 . 1** Discuss the reasons why the conservation of biodiversity is important for human society.

**[9 marks]**

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**1 0 . 2** State what is meant by **biodiversity**.

**[1 mark]**

**10**

**Write an essay on one of the following topics.**

**1 1 . 1** Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

**[25 marks]**

**OR**

**1 1 . 2** Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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**END OF QUESTIONS**